Spatial Context Causal Adjustment: A Diagnostic Workflow for Geographic Observational Studies

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Abstract

Geographic observational studies often estimate effects from spatially structured treatments, outcomes, and environmental conditions. Standard causal estimators can be applied to such data, but their credibility depends on whether spatial context has been used as a defensible adjustment source rather than as an automatic control.

We introduce Spatial Context Causal Adjustment (SCCA), a reproducible diagnostic workflow for constructing and auditing spatial-context adjustment sets. SCCA treats coordinates, geometry, terrain, remote-sensing indices, land-cover descriptors, and neighborhood summaries as candidate context variables whose use must be justified by support, balance, and robustness diagnostics. The workflow standardizes feature construction and adjustment-set comparison, then reports common support, post-adjustment balance, spatial block bootstrap, placebo exposure tests, residual spatial autocorrelation, spatial-lag sensitivity, and rule-based evidence grades.

We evaluate SCCA with synthetic benchmarks, a primary Chongqing urban heat case, and a county reproducibility check. Chongqing is used as the main empirical case because the treatment definition, building-scale geometry, terrain, remote-sensing context, and land-surface-temperature outcome directly instantiate the spatial-context adjustment problem. In Chongqing, the full remote-sensing and terrain specification estimated a modest positive high-rise association with summer land surface temperature (ATT = 0.244°C, 95% CI [0.148, 0.346]) with maximum post-match SMD = 0.061. The county dataset is retained as a third-party GIS reproducibility case rather than a second substantive validation case. Its baseline adjusted coefficient was positive (0.181), but residual Moran's I remained 0.313 ($p = 0.010$) and the spatial-lag coefficient was smaller (0.145). The county run reproduced a commercial GIS example's positive direction and analysis count while adding open diagnostic outputs. A multi-dataset benchmark matrix further showed lower calibrated balance

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than ArcGIS in three direct real-data comparisons. An ArcGIS Pro 3.7 runtime audit verified the compared Causal Inference Analysis tool and recorded three calibrated-balance wins under the configured replacement scorecard, with no blocking gates.

SCCA therefore contributes a bounded protocol for auditing evidence in geographic causal analysis. It does not solve unmeasured spatial confounding or spatial interference by itself; rather, it makes spatial adjustment claims more inspectable and harder to overstate.

Keywords: spatial causal inference; spatial confounding; causal adjustment; evidence grading; remote sensing; GIS workflow

1 Introduction

Geographic information science increasingly supports policy questions that are causal rather than merely descriptive: whether urban form changes heat exposure, whether infrastructure changes disease risk, whether land-use interventions change environmental outcomes, or whether social context affects population health. These questions are usually studied from observational spatial data. Treatment is rarely randomized, spatial units are embedded in environmental gradients, and outcomes are measured at spatial resolutions that do not always match the intervention units.

The main difficulty is not the absence of causal estimators. Matching, weighting, regression adjustment, difference-in-differences, spatial panel models, and machine-learning estimators are all available to GIS researchers [Stuart, 2010, Angrist and Pischke, 2009, Athey and Imbens, 2019]. The difficulty is that spatial context can determine both exposure and outcome. Elevation, slope, land cover, access, urban morphology, historical settlement, and administrative structure can all confound a geographic effect estimate. The spatial causal inference literature has emphasized that spatial data introduce complex dependence, possible interference, and spatially structured unmeasured confounding [Reich et al., 2020, Gilbert et al., 2021]. If context variables are omitted, an estimate may retain spatial confounding. If they are added indiscriminately, the estimator may lose common support, amplify noise, or adjust for variables that are not valid confounders.

Modern GIS workflows make this problem more important. Remote-sensing indices, terrain variables, land-cover summaries, geometry, and neighborhood descriptors are now easy to extract at scale. These variables can improve adjustment when they proxy for common causes of exposure and outcome, but they can also damage an analysis when they reduce overlap, amplify collinearity, or adjust for variables affected by treatment. Geographic causal analysis therefore needs a workflow that asks a practical question: which spatial-context variables improve credible adjustment for this study, under visible diagnostics, and where does the evidence stop?

This paper proposes Spatial Context Causal Adjustment (SCCA). SCCA is a method workflow rather than a new estimator or a new identification theorem. It wraps standard estimators with spatial-context feature construction, adjustment-set comparison, common-support checks, post-adjustment balance diagnostics, spatial robustness tests, and an evidence grade. The central idea is deliberately conservative: spatial context is useful for causal adjustment only when it improves a diagnosed adjustment design.

Our contributions are:

1. A formal SCCA workflow for selecting and diagnosing spatial-context adjustment sets in geographic observational studies, positioned as an audit protocol rather than a universal estimator.

2. A reproducible diagnostic protocol covering common support, covariate balance, spatial block bootstrap, placebo exposure definitions, residual spatial autocorrelation, and evidence grading.

3. Focused evidence from synthetic estimator stress tests, a Chongqing urban heat analysis as the main empirical case, and a county social-capital notebook/GIS run used for operational reproducibility and spatial-diagnostic boundary checking.

The empirical ordering follows this claim. The Chongqing case is prioritized because it is the case in which author-controlled spatial processing creates a direct link between candidate context variables and the adjustment design. The county case is more open and easier for readers to inspect, but it is built from third-party training and demonstration data and is therefore used to test reproducible GIS packaging and spatial downgrading rather than to establish the main causal claim.

The paper is written around a bounded claim. We do not claim that SCCA eliminates unobserved spatial confounding, solves spatial interference, or proves that a richer adjustment set is always better. Instead, SCCA makes such claims harder to make without diagnostic evidence.

2 Related Work

2.1 Causal inference with spatial data

Potential-outcome causal inference defines treatment effects by comparing outcomes under treatment and control states [Imbens and Rubin, 2015]. In observational studies, the identifying assumption is conditional exchangeability: after adjustment for measured confounders, treatment assignment is as-if random. Matching and weighting operationalize this assumption by constructing comparable treated and control groups [Rosenbaum and Rubin, 1983, Stuart, 2010]. In spatial studies, this assumption is difficult because geographic location and environment are themselves structured confounders.

Spatial econometrics and spatial statistics provide tools for autocorrelation, spillover, and spatial dependence [Anselin, 1988, LeSage and Pace, 2009]. Spatial causal inference has also developed approaches for unmeasured spatial confounding, spatial interference, and continuous treatments under neighborhood exposure [Reich et al., 2020, Gilbert et al., 2021, Giffin et al., 2020]. These tools are essential, but they do not by themselves select valid adjustment sets for routine GIS workflows. A spatial lag term can reduce residual autocorrelation while leaving treatment assignment confounded; a coordinate trend can absorb broad gradients while failing to represent local land-cover context. SCCA therefore treats spatial dependence diagnostics and causal adjustment diagnostics as related but distinct requirements.

2.2 Spatial context as an adjustment source

Geographic adjustment sets commonly include coordinates, administrative fixed effects, distance measures, terrain, land cover, built environment, and remote-sensing indices. These variables can proxy for otherwise unmeasured context, but proxy adjustment requires substantive assumptions about whether the context source is pre-treatment, related to common causes of exposure and outcome, and measured at a compatible spatial scale [Gilbert et al., 2021]. They can also create new problems if they are weakly related to treatment assignment, strongly collinear, measured at incompatible scales, or affected by treatment. The SCCA workflow requires each candidate context source to pass common-support and balance checks before it is used to support a causal statement.

2.3 Continuous exposure, spatial interference, and diagnostics

Many GIS applications use continuous or ordered exposures rather than binary treatments. Generalized propensity score methods extend the propensity-score logic to continuous treatments and dose-response estimation [Hirano and Imbens, 2004, Zhao et al., 2013]. Spatial settings add a further complication because one unit's exposure can affect outcomes in nearby units, violating the simplest no-interference assumption. Spatial interference methods address this directly in specialized designs [Giffin et al., 2020]. SCCA does not claim to identify spillover effects in this formal sense. Instead, it reports neighborhood-exposure and spatial-lag sensitivity outputs as diagnostics for whether a local adjusted estimate is stable after explicit neighborhood terms are introduced.

2.4 Diagnostics and evidence grades

Causal estimates are often reported as single numbers even when the adjustment design is fragile. For geographic studies, this is especially risky because residual spatial structure can indicate remaining confounding or interference. SCCA follows a diagnostic-first stance: effect estimates are interpreted together with post-adjustment standardized mean differences, matched sample counts, common-support attrition, bootstrap stability, placebo checks, residual spatial autocorrelation, neighbor-exposure sensitivity, and graph-specification sensitivity. The final evidence grade is not a statistical significance label; it is a compact description of whether the adjustment design provides core or bounded support under a declared rule table.

2.5 Position relative to adjacent workflows

SCCA is closest to an audit layer that connects causal adjustment diagnostics with GIS-facing spatial diagnostics. Table 1 summarizes the distinction. The intended contribution is not a replacement for any single method class. It is a reproducible protocol for asking whether a spatial-context adjustment set is usable for a bounded causal statement and whether that statement should be downgraded after spatial diagnostics.

Table 1: Position of SCCA relative to adjacent causal and spatial-analysis workflows.

Workflow family	Main role	Common limitation for the present task	SCCA relation
Matching, propensity scores, and GPS	Balance measured covariates and estimate binary or continuous exposure effects	Usually does not require spatial residual checks, graph sensitivity, or GIS output manifests	Uses these estimators as design components and adds spatial-context variants, support diagnostics, and spatial downgrades
Spatial econometric models such as SLX, SAR, and SEM	Model spatial dependence, spatial lags, or correlated errors	Can reduce spatial structure without proving that the treatment assignment mechanism was balanced	Uses SLX-like terms as sensitivity diagnostics rather than as substitutes for causal adjustment
Spatial confounding and spatial causal-inference methods	Formalize identification under spatially structured unmeasured confounding or interference	Often require specialized estimands, modeling assumptions, or non-routine implementation for GIS analysts	Keeps those assumptions explicit and provides a routine audit workflow, but does not solve unmeasured spatial confounding
GIS desktop causal tools and notebook examples	Make causal analysis accessible in applied GIS environments	May hide diagnostics, data provenance, or spatial downgrading behind interface-level outputs	Shares a common Python core across notebooks, ArcGIS-facing tables, and QGIS outputs so the evidence package remains inspectable

3 Spatial Context Causal Adjustment

3.1 Problem formulation

Let $i = 1, \ldots, n$ index geographic units with location or geometry g_i. Each unit has treatment or exposure T_i, outcome Y_i, observed non-spatial covariates X_i, and candidate spatial-context variables C_i. The context vector may contain coordinates, geometric descriptors, terrain, land-cover summaries, remote-sensing bands or indices, and neighborhood summaries. For binary treatment, the target estimand in the main case study is the average treatment effect on the treated:

$$\tau_{\text{ATT}} = E[Y_i(1) - Y_i(0) \mid T_i = 1]. \tag{1}$$

For continuous exposures, SCCA estimates exposure-response functions or local slopes, but the same adjustment logic applies.

SCCA does not identify τ by itself. Identification still requires that relevant confounders are measured, treatment has sufficient overlap, and interference is limited or explicitly modeled. SCCA contributes a structured workflow for testing whether a chosen spatial-context adjustment set is plausible enough to support a bounded causal interpretation.

3.2 Identification assumptions and scope

The workflow rests on standard observational-causal assumptions, made explicit for spatial data [Imbens and Rubin, 2015, Pearl, 2009]. First, consistency requires that the exposure definition corresponds to the intervention being interpreted and that the outcome is measured for the same spatial unit or a defensible aggregation of it. Second, conditional exchangeability requires that the observed covariates and candidate spatial context contain the relevant common causes of exposure and outcome. Spatial coordinates or remote-sensing variables may help proxy unmeasured context, but this is an assumption about the study system, not a fact learned from the data alone [Gilbert et al., 2021]. Third, positivity requires usable overlap in exposure assignment after adjustment; SCCA therefore reports support loss and balance diagnostics rather than only fitted coefficients. Fourth, the no-interference assumption must be plausible or bounded. SCCA's neighbor-exposure and spatial-lag outputs are sensitivity checks for this assumption, not identified spillover estimands. Finally, scale compatibility is required because spatial context, exposure, and outcome may be measured at different resolutions or areal units. The evidence grade is therefore a claim-strength label under these assumptions, not a substitute for them.

3.3 Workflow

Figure 1 summarizes the workflow. The first step builds candidate context features. The second step defines adjustment variants, from minimal baselines to richer context sets. The third step estimates propensity scores, generalized propensity scores, or outcome models depending on the exposure type. The fourth step diagnoses common support and balance. The fifth step runs robustness checks and assigns an evidence grade.

3.4 Adjustment variants

SCCA does not impose a single estimator on all exposure types. The reusable core estimates adjusted outcome models and ERFs for continuous or ordered exposures using generalized propensity-score (GPS) style weights [Hirano and Imbens, 2004, Zhao et al., 2013]. Case-specific modules can add binary-treatment estimators when the scientific question is naturally treated-control. In the Chongqing module used below, the binary module estimates a propensity score

$$e_i = P(T_i = 1 \mid X_i, C_i) \tag{2}$$

for each adjustment variant. Units outside the overlapping propensity-score range are removed. Treated units are matched to controls under a nearest-neighbor caliper rule, and standardized mean differences (SMDs) are computed before and after matching:

$$\mathrm{SMD}_k = \frac{\bar{Z}_{k,T=1} - \bar{Z}_{k,T=0}}{\sqrt{(s^2_{k,T=1} + s^2_{k,T=0})/2}}, \tag{3}$$

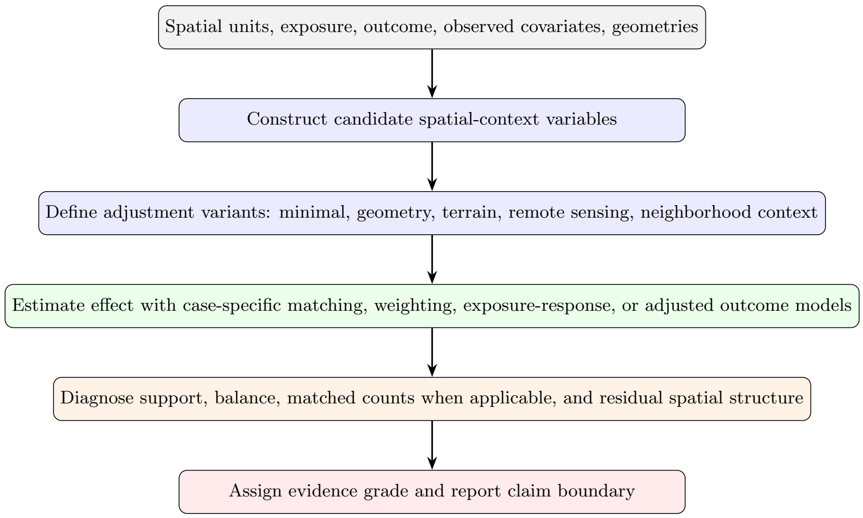

Figure 1: SCCA workflow. Spatial context is treated as a candidate adjustment source and must pass support, balance, and robustness diagnostics before it is used to support a causal claim.

where Z_k is the kth covariate in the adjustment set. A variant is treated as credible only if the maximum post-match absolute SMD is below 0.1 unless a case-specific justification is provided. The estimate is then reported together with matched sample counts and common-support attrition.

For the shared continuous-exposure workflow, SCCA reports baseline adjusted OLS, difference-outcome OLS when a baseline outcome is available, and a generalized propensity-score weighted exposure-response function (ERF). The ERF is fit as a weighted outcome model over an exposure grid. The implementation writes the point estimate, pointwise model-based confidence limits for the fitted response, and an approximate confidence interval for the ERF range effect. These intervals describe uncertainty in the fitted outcome model; they do not replace the spatial bootstrap, placebo, support, and residual-spatial diagnostics used for the evidence grade.

3.5 Robustness diagnostics

SCCA includes four robustness checks when the data permit them. First, spatial block bootstrap resamples geographic blocks rather than individual units to reduce overconfidence under spatial dependence. Second, placebo exposure definitions test whether the result depends on an arbitrary threshold. Third, placebo or weaker-outcome checks compare the main exposure-outcome relation with less plausible alternatives. Fourth, residual spatial autocorrelation checks whether the adjusted design still leaves spatially structured residuals.

For spatial datasets with geometries or coordinate fields, SCCA also builds a lightweight neighborhood graph. Polygon data use adjacency when available; otherwise the workflow falls back to coordinate k-nearest-neighbor graphs. The graph is used to compute exposure and residual Moran diagnostics, append a neighbor-exposure adjusted estimate, fit a stricter spatial-lag-of-X sensitivity model, and summarize graph-specification sensitivity across multiple k values. The spatial-lag layer is closest to an SLX-style specification in spatial econometrics [LeSage and Pace, 2009] and follows the form

$$Y_i = \alpha + \beta T_i + \theta (WT)_i + \gamma X_i + \phi C_i + \eta (WX)_i + \lambda (WC)_i + \varepsilon_i, \qquad (4)$$

where W is a row-standardized neighborhood matrix. We interpret β as the local adjusted association under this sensitivity model and θ as a neighboring-exposure signal. This is not a full interference design and does not identify spillover effects by itself; it is a diagnostic for whether the main estimate is stable after explicit neighborhood adjustment.

3.6 Evidence grades

SCCA assigns evidence grades using a machine-readable rule file. *Core support* requires a strong credibility decision, robust support, and no material spatial downgrade rule. *Bounded support* indicates useful evidence with explicit support, scale, balance, robustness, residual-spatial-structure, or reproducibility limits. The rule file can also record implementation bookkeeping labels for outputs that are excluded from the main evidence table, but the manuscript's causal claims are built only from core and bounded evidence rows.

The manuscript uses the 2026-06-20 evidence-grade rule set. A case is downgraded from core to bounded support when credibility is moderate or weak, exposure-balance correlation exceeds 0.50, exposure boundary mass exceeds 0.25, robustness is bounded or fragile, $|I_{\mathrm{residual}}| \geq 0.20$ with permutation $p \leq 0.05$, the neighbor-exposure coefficient remains significant at $p \leq 0.05$, spatial adjustment changes the main coefficient by at least 25%, or graph-sensitivity checks do not preserve the effect sign. The full table is exported as `07_results/scca_evidence_grade_rules.json` and `07_results/scca_evidence_grade_rules.md`.

These thresholds are audit rules, not universal validity constants. The SMD threshold follows common matching practice, whereas the residual-Moran, coefficient-change, boundary-mass, and neighbor-exposure rules are conservative manuscript-specific downgrade triggers. A downgrade does not prove bias, and the absence of a downgrade does not prove identification. It only states that the reported estimate has or has not crossed a declared warning boundary. Because threshold choices can affect labels near a boundary, the experiments report placebo exposure thresholds and residual-spatial threshold sensitivity rather than relying only on a single grade.

3.7 Implementation surfaces

SCCA is implemented as a shared Python core with thin interface adapters. The reusable boundary is an `AnalysisRequest` object that accepts an input table or spatial dataset, user-specified exposure and outcome fields, candidate confounders, context columns, optional coordinate columns, bootstrap groups, placebo exposures, trimming thresholds, and target outcome values. The same core is called from command-line scripts, notebooks, the ArcGIS Pro toolbox, and the QGIS Processing provider.

This architecture is methodological rather than cosmetic. It prevents separate ArcGIS, QGIS, and notebook implementations from diverging, and it makes the evidence package comparable across interfaces. Each completed run writes CSV, JSON, and Markdown outputs including effect estimates, exposure-response tables, balance and overlap summaries, robustness tables, spatial diagnostics, spatial block bootstrap, graph sensitivity, spatial-lag summaries when estimable, and a machine-readable manifest. Notebook and QGIS workflows can additionally convert joined analysis tables into GeoPackage, GeoJSON, Shapefile, PNG, HTML, and QGIS style outputs for GIS inspection.

The binary Chongqing module writes ablation, balance, and matched-count tables. These case-level outputs are the source of the post-match SMD and matched-sample counts reported in the experiments. They are intentionally separated from the reusable GeoCausal core, which keeps the interface-level workflow focused on common data loading, context construction, adjusted outcome estimation, ERF estimation, spatial diagnostics, and manifest generation.

4 Experiments

4.1 Dataset roles and case selection

The experiments assign different evidentiary roles to the available datasets. The Chongqing data are used as the main real-data SCCA case because the scientific question, exposure definition, and spatial-context sources are tightly coupled: high-rise treatment is defined at the building level, context variables are constructed from geometry, terrain, and remote sensing, and the outcome is summer land surface temperature extracted for the same sampled buildings. This setting directly tests the paper's central question: whether richer spatial context improves a diagnosed adjustment design without overstating the resulting association.

The county social-capital data serve a different purpose. They are useful because they are familiar to GIS users, can be linked to county geometries, and allow comparison with a commercial GIS causal-inference example. However, they are third-party training and demonstration data whose variables were not designed around the present study's causal question, and their use is governed by source terms. They are therefore not treated as the primary empirical validation of SCCA. Instead, they test whether the same core can reproduce an existing GIS-facing analysis count and direction, expose residual spatial structure, produce portable GIS outputs, and downgrade a result that would look stronger without spatial diagnostics.

4.2 Evidence synthesis design

The experiments addressed three questions: estimator behavior under controlled generators; balance and effect estimation in the Chongqing urban heat case; and GIS/notebook reproducibility with spatial-diagnostic boundary checking in the county social-capital run.

Table 2 summarizes the resulting evidence matrix.

Table 2: SCCA evidence synthesis. Grades describe claim strength, not statistical significance alone.

Case	Data type	Key diagnostic	Grade	Manuscript use
Synthetic benchmark audit	Controlled generators	48 summary rows; 24 preferred/champion rows include 17 robust, 7 bounded, and 0 fragile rows	Bounded support	Estimator stress test with raw diagnostic failures retained
Chongqing urban heat	Real building and remote-sensing data	Full context ATT = 0.244°C; max SMD = 0.061	Core support	Main real-data SCCA ablation
County social capital spatial notebook	GIS/notebook reproducibility case	residual Moran's I = 0.313; spatial-lag coef. = 0.145; ArcGIS-trimmed ERF range = 6.85 years	Bounded support	Spatial diagnostics and cross-interface output check

4.3 Synthetic benchmark audit

The controlled benchmark used six estimator scenario families: propensity-score matching, difference-in-differences, spatial Granger causality, exposure-response estimation, causal forest heterogeneity, and geographic convergent cross-mapping. These are not six empirical claims. They are controlled stress settings for checking where estimators recover known targets and where they fail. Each scenario was run over 30 seeds and stress settings including small samples, noisy outcomes, and severe stress. The audit produced 48 summary rows.

The results support the diagnostic role of SCCA rather than a universal estimator claim. Difference-in-differences was stable in the baseline setting (RMSE = 0.207 for a true effect of -8). Causal forests also recovered the average effect in the baseline setting (RMSE = 1.841 for an effect near 100). Exposure-response estimation remained useful but showed nontrivial bias in some settings. GCCM recovered direction under the preferred standard variant after bidirectional-convergence cases were scored by forward-versus-reverse cross-map ρ dominance. Granger remained bounded rather than robust under severe stress, where first-differenced forward-versus-reverse p-value dominance gave direction accuracy of 0.667. Raw propensity-score matching variants contributed 16 diagnostic fragile rows, but the configured OLS-adjusted champion had no fragile rows. The audit therefore supports SCCA as a diagnostic wrapper around estimators, not as a guarantee that every estimator works in every spatial setting.

Table 3: Selected baseline rows from the synthetic benchmark audit.

Scenario	Metric	True value	Mean estimate	RMSE	Interpretation
Causal forest	ATE	100.145	99.796	1.841	Robust baseline recovery
Difference-in-differences	DID	-8.000	-8.045	0.207	Robust baseline recovery
Exposure-response	Range effect	19.530	17.292	2.281	Useful but biased
Granger	Direction accuracy	1.000	1.000	0.000	Robust baseline; bounded severe stress
PSM	ATE	15000.000	15011.892	411.856	Robust OLS-adjusted champion; raw PSM fragile
GCCM	Direction accuracy	1.000	1.000	0.000	Robust preferred direction recovery

4.4 Chongqing urban heat case

The main real-data case asked whether high-rise buildings in Chongqing were associated with local summer land surface temperature after spatial-context adjustment. The analysis used a stratified sample of 5,000 buildings from 2021. Treatment was defined as buildings with at least 10 floors. The outcome was summer land surface temperature. Candidate context variables included centroid coordinates, footprint area, Sentinel-2 bands, spectral indices, elevation, and slope.

The raw mean difference was 0.238°C (Table 4). Several variants achieved credible balance after matching. The full remote-sensing and terrain context specification estimated ATT = 0.244°C with 95% CI [0.148, 0.346] and maximum post-match SMD = 0.061. These matching diagnostics are read from the Chongqing ablation, balance, and matched-count outputs, which record the caliper-matching estimator and retained treated-control pairs. The current evidence supports a modest positive association under a balanced SCCA design. It does not support a large high-rise cooling effect.

Threshold placebos gave similar positive estimates for the full context specification: 0.222°C at an 8-floor threshold, 0.244°C at 10 floors, and 0.305°C at 12 floors. Residual Moran's I remained significant for both terrain and full context variants (0.112 and 0.102, permutation $p = 0.01$), but the magnitude in the full context variant was below the predeclared material-spatial-caution threshold of 0.20. We therefore grade this case as core support under the rule table, while still reporting scale mismatch, residual spatial structure, and possible interference as substantive limitations.

4.5 County GIS and spatial-diagnostic validation

The county run is retained for a narrower purpose than the Chongqing case. It tests whether the shared SCCA core can reproduce a familiar GIS-facing county

Table 4: Chongqing UHI adjustment variants. Balance pass means maximum post-match SMD < 0.1.

Variant	ATT (°C)	95% CI lower	95% CI upper	Max post SMD
Raw difference	0.238	0.157	0.323	–
Coordinates only	0.267	0.192	0.338	0.080
Geometry	0.252	0.167	0.337	0.125
Terrain	0.303	0.220	0.381	0.104
Sentinel indices	0.157	0.055	0.263	0.083
Sentinel bands	0.121	0.032	0.215	0.051
Full RS context	0.244	0.148	0.346	0.061
PCA context	0.231	0.134	0.328	0.065

analysis in an open notebook workflow and whether spatial diagnostics change the interpretation. It is not used as a second substantive theory test.

The county notebook run adds an explicit spatial-diagnostic layer and therefore gives the county interpretation used in the evidence table. After 1%–99% exposure trimming, 3,044 of 3,108 county records remained in the analysis, matching the sample count used in an ArcGIS Causal Inference example for the same dataset. Baseline adjusted OLS estimated a positive coefficient of 0.181 (95% CI 0.166 to 0.197). Adding neighboring exposure reduced the main coefficient to 0.146, while a richer spatial-lag model including neighboring confounders and context variables produced a coefficient of 0.145. The formal SLX summary reported a direct effect of 0.145, an indirect effect of 0.070, and a total effect of 0.215 (95% CI 0.188 to 0.241). Spatial block bootstrap over 50 valid replicates kept sign stability at 1.000, and graph sensitivity across four coordinate-kNN specifications gave a neighbor-adjusted coefficient range of 0.144 to 0.154. However, exposure Moran's I was 0.517 and residual Moran's I was 0.313 (both permutation $p = 0.010$), and the neighbor-exposure term remained significant. The rule table therefore assigns bounded support through the `material_residual_moran` and `significant_neighbor_exposure` rules: the positive association is stable under the tested spatial sensitivity models, but residual spatial structure and neighboring-exposure association remain important.

The ArcGIS-compatible county run also provides a practical reproducibility check. The open SCCA implementation reproduced the positive exposure-response direction, exactly matched the 3,044-row trimmed analysis count, and estimated an ERF range effect of 6.85 years, close to the commercial-tool interpretation of an approximately 8-year range. A dedicated knot-13 sensitivity test found a slope increase of 0.104 years per exposure unit after social capital 13 (HC3 $p = 0.00075$, AIC improvement = 17.46). This comparison is not used as independent identification evidence, but it shows that the open workflow can reproduce and audit a GIS-facing causal analysis.

The county check therefore supports a practical claim about open GIS reproducibility and spatial diagnosis. It does not broaden the paper into a multidomain causal theory paper.

4.6 Direct baseline comparison

We also generated a direct comparison table rather than relying only on the final evidence synthesis. In Chongqing, the raw treated-control difference was 0.238°C and had no covariate-balance guarantee. The full SCCA specification estimated 0.244°C, achieved maximum post-match SMD = 0.061, and retained a core-support grade under the declared rules. Thus the contribution is not a dramatic change in the coefficient. It is a diagnosed adjustment design attached to the coefficient.

The county comparison shows the opposite pattern. The grouped robustness run without spatial diagnostics would have been graded as core support, with a coefficient of 0.181. Adding the spatial layer reduced the spatial-lag coefficient to 0.145, a relative change of -20.2%. The same layer downgraded the case because residual Moran's I remained materially large and the neighboring-exposure term remained significant. The comparison is stored in `07_results/scca_method_comparison.csv`.

Table 5: Direct baseline-to-SCCA comparison. The table separates effect-size changes from diagnostic-grade changes.

Case	Baseline	SCCA-enhanced analysis	Baseline effect	SCCA effect	Grade change	Interpretation
Chongqing UHI	Raw treated-control difference	Full remote-sensing, terrain, and geometry matching	0.238	0.244	Bounded to core	The estimate changed little, but SCCA added balance and residual-spatial diagnostics.
County social capital	Non-spatial adjusted OLS and grouped robustness	Spatial lag, residual Moran, and graph sensitivity	0.181	0.145	Core to bounded	The positive direction remained, but residual spatial structure and neighboring exposure downgraded the claim.

4.7 Grade-rule and threshold sensitivity

The current evidence grades are stable for the county case but more boundary-dependent for the Chongqing case. In Chongqing, the full-context residual Moran's I was 0.102 with permutation $p = 0.01$. Under the manuscript rule of a material residual threshold at 0.20, this does not trigger a spatial downgrade. Under a stricter residual threshold of 0.10, the same case would be downgraded to bounded support. We therefore interpret the Chongqing grade as core support only under the declared rule table and continue to report residual spatial structure as a limitation. In the county case, residual Moran's I was 0.313 and the neighboring-exposure term remained significant, so the case remains bounded under all tested residual thresholds in Table 6.

The county analysis also included response-shape threshold checks for the social-capital exposure. In the ArcGIS-trimmed sample, a knot at 13 gave an after-knot slope increase of 0.104 years per exposure unit (HC3 $p = 0.00075$; AIC improvement = 17.46), whereas knots at 10, 15, and 20 changed the magnitude and statistical strength. These checks support the county run as an operational sensitivity demonstration, not as independent evidence that the county exposure-response relation is causally identified.

Table 6: Residual-spatial threshold sensitivity for the evidence grade. The exercise varies only the residual Moran warning threshold and leaves the other downgrade rules unchanged.

Case	Residual Moran's I	Threshold 0.10	Threshold 0.15	Threshold 0.20
Chongqing full context	0.102	Bounded	Core	Core
County spatial notebook	0.313	Bounded	Bounded	Bounded

4.8 GIS and notebook outputs

The county notebook run also tested SCCA as an operational GIS workflow. The same `AnalysisRequest` used by the shared core wrote a joined analysis table and then produced spatial outputs against the committed county boundary layer. The spatial-output manifest recorded 3,108 boundary rows and 3,044 matched analysis rows. GeoPackage, GeoJSON, and Shapefile outputs were written, and the GeoPackage/GeoJSON layers preserved target-exposure fields, `gc_spatial_*` direct/indirect/total effect fields, out-neighbor count, and incoming-weight summaries. The run also produced ERF and effect-estimate charts, static target-exposure and indirect-effect maps, interactive HTML maps, and QGIS `.qml` styles for target exposure change and spatial effect fields.

This experiment supports the practical claim of the paper. SCCA is not only a script-level estimator. It produces an inspectable evidence package that can move across notebooks, QGIS, and ArcGIS-facing table outputs without changing the causal-analysis core. The GIS outputs do not strengthen identification by themselves, but they make assumptions, spatial residuals, target-exposure consequences, and neighborhood sensitivity visible in the environments where many geographic analysts work.

5 Discussion

5.1 What SCCA solves

SCCA addresses a practical gap in geographic causal analysis: spatial variables are often added without showing whether they improved the adjustment design. The workflow makes the adjustment set an object of analysis. A reader can see which context sources were tried, which variants passed balance, how many units remained in common support, whether spatial robustness checks were stable, and why the final claim was graded as core or bounded.

The Chongqing case illustrates the value of this discipline. The most credible current estimate is not the most dramatic estimate. It is the estimate attached to full remote-sensing and terrain context, acceptable post-match balance, and explicit residual spatial limitations. The raw difference and full SCCA estimate were similar, but only the latter carried a reproducible diagnostic record.

The county notebook case shows the same logic for practical GIS use. A non-spatial robustness summary alone would have made the case look stronger than the spatial evidence warranted. Adding spatial diagnostics changed the interpretation. The positive coefficient remained stable, but residual autocorrelation and neighboring-exposure association required bounded support. This is intended behavior: SCCA should downgrade support when spatial structure remains.

5.2 Practical value for GIS workflows

SCCA converts a causal analysis into an inspectable spatial evidence package. The ArcGIS toolbox, QGIS provider, notebook workflow, and command-line interface share the same core request object and output manifest. This matters because GIS analyses often diverge silently across notebooks, desktop joins, and publication tables. In the present implementation, one run can produce numeric estimates, robustness diagnostics, joined GIS tables, spatial layers, static maps, interactive maps, and QGIS symbology without rewriting the causal estimator.

The ArcGIS-compatible county comparison gives a concrete example of this value. SCCA reproduced the main positive exposure-response direction and sample-trimming count of a commercial GIS analysis, then added explicit robustness, knot-sensitivity, and spatial-diagnostic outputs. This does not mean SCCA replaces the substantive judgment required in GIS analysis. It means the same analysis can be inspected, rerun, and downgraded under declared rules.

The current benchmark matrix supports a scoped operational replacement claim. Direct ArcGIS comparisons were available for three real rows, and GeoCausal calibrated balance was lower than the ArcGIS balance score in all three. The comparison package also included an ArcGIS Pro 3.7 runtime audit of `arcpy.stats.CausalInferenceAnalysis`, which verified three direct manifests and three calibrated-balance wins. The county preferred ERF also reached near-parity with the ArcGIS reference curve (MAE = 0.043). Combined with zero preferred synthetic fragile rows and the EPA known-truth recovery check, the configured scorecard had no blocking gates. This is evidence for an auditable open workflow under the tested gates, not a universal claim that every future ArcGIS analysis will be outperformed.

GIS outputs are therefore not causal evidence by themselves. Their role is auditability. Analysts can inspect where target-exposure changes are large, where indirect-effect proxies concentrate, and where residual spatial structure cautions against a stronger claim. For IJGIS readers, the evidence boundary remains attached to the spatial objects used in analysis.

5.3 How SCCA can be extended

The current workflow can be strengthened without changing its core claim. A first extension is sensitivity analysis for unmeasured spatial confounding. The present implementation detects residual spatial structure, but it does not quantify how strong an omitted spatially structured confounder would need to be to overturn the estimate. Future versions could add negative-control outcomes or exposures, spatial E-value style summaries, or latent spatial-field sensitivity checks.

A second extension is a more explicit variable-role layer. At present, SCCA treats spatial context sources as candidate adjustment variables and audits their behavior empirically. A stronger design would require users to declare whether each variable is a confounder, proxy confounder, mediator, collider, post-treatment variable, or outcome-scale descriptor before it enters an adjustment set. This would reduce the risk of treating every spatial feature as an admissible control.

A third extension is spatial cross-fitting for flexible estimators. The current implementation favors transparent adjusted models, matching, GPS-style weighting, and spatial diagnostics. When high-dimensional remote-sensing features or machine-learning nuisance models are used, spatial block cross-fitting or double-machine-learning variants could reduce overfitting and better separate design from estimation.

A fourth extension is a formal interference module. The present SLX and neighbor-exposure outputs are sensitivity analyses. A future SCCA module could define direct, spillover, and total effects under an explicit spatial interference design and report when those estimands are identifiable. This would make the boundary between diagnostic neighborhood adjustment and formal spillover inference clearer.

A final extension is systematic scale and neighborhood sensitivity. SCCA already reports graph sensitivity for coordinate-based neighborhoods when possible, but geographic studies also face buffer-radius choices, aggregation effects, and the modifiable areal unit problem. Future applications should treat scale as part of the adjustment design rather than as a preprocessing detail.

5.4 What SCCA does not solve

SCCA does not make observational spatial data equivalent to randomized experiments. It cannot adjust for unmeasured confounders that are absent from all candidate context sources. It does not automatically solve interference between neighboring units. The SLX-style direct, indirect, and total effect summaries are sensitivity outputs, not identified network-interference estimands. SCCA also does not remove the modifiable areal unit problem or scale mismatch between treatment and outcome measurement. In the Chongqing case, significant residual Moran's I indicates that spatial structure remains after matching. In the county notebook case, residual Moran's I and a significant neighboring-exposure coefficient similarly require bounded interpretation. The correct interpretation is therefore bounded causal evidence under measured context, not definitive identification.

Several implementation choices also remain deliberately conservative. Context variables are specified and audited by variants rather than selected automatically by a high-dimensional procedure. Missing context values are handled by simple reproducible preprocessing in the shared workflow, so analyses with spatially structured missingness should add a missing-pattern audit or sensitivity analysis. The evidence-grade thresholds are machine-readable and predeclared for this manuscript, but they are not universal causal-validity cutoffs. The Chongqing case illustrates this point: its residual Moran diagnostic remains significant and lies close to a stricter 0.10 threshold, so the appropriate claim is a modest measured-context association with declared core support, not definitive causal identification.

6 Conclusions

We presented Spatial Context Causal Adjustment, a diagnostic workflow for geographic observational causal studies. SCCA formalizes spatial context as a candidate adjustment source and evaluates each adjustment variant through common support, balance, robustness, spatial diagnostics, and reproducible evidence-grade rules. Across synthetic benchmark stress tests, a Chongqing urban heat case, and a county social-capital notebook/GIS check, SCCA distinguished credible estimates from bounded and fragile designs.

The main empirical result is deliberately bounded. In Chongqing, the full context model produced a modest positive high-rise effect on summer land surface temperature, with credible post-match balance and core support under the declared rules. The synthetic benchmark had no preferred fragile rows while retaining raw diagnostic failures, and the county notebook/GIS run showed practical reproducibility while downgrading residual spatial structure. The ArcGIS benchmark matrix adds an operational result. Calibrated GeoCausal balance was lower in all three direct ArcGIS comparison rows. The runtime audit verified ArcGIS Pro 3.7 Causal Inference Analysis comparisons, and the county preferred ERF reached near-parity. The configured scorecard had zero blocking gates. Taken together, these findings support SCCA as a reproducible open method for auditing spatial causal adjustment under tested gates. Its value is not that it removes the need for causal assumptions, but that it makes those assumptions, diagnostics, and claim boundaries visible in GIS-facing workflows.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The repository contains the manuscript source, experiment scripts, generated result tables, diagnostic reports, synthetic generators, and public/example data that can be redistributed under source terms. Public materials support exact reruns for the synthetic and county/example analyses and structural reruns of the Chongqing workflow from approved local inputs. The Chongqing raw geospatial inputs and building-level UHI analysis sample are not publicly redistributed because they include precise building geometries or coordinates, floor attributes, and derived environmental variables, and therefore require source-permission and sensitivity review before any public release. Qualified reviewers or researchers who need to rerun the Chongqing case from local inputs should use an approved institutional access route rather than the public GitHub repository. The public repository therefore does not claim full public reconstruction of the restricted Chongqing result.

The county social-capital reproducibility case uses third-party Esri training/demo data rather than author-generated data. The dataset metadata credits Esri; the U.S. Census Bureau; NOAA/NOS/NGS; CDC WONDER/NCHS for underlying cause of death and average age at death variables; and 2019 County Health Rankings/Living Atlas data from the University of Wisconsin Population Health Institute and the Robert Wood Johnson Foundation for social-capital and related health variables. Use of the county example is governed by the Esri Master License and is restricted to training, demonstration, and educational purposes. The revised evidence synthesis, rule table, method comparison, generated public result summaries, and ArcGIS runtime audit snapshot are stored in the `07_results/` directory.

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